

1) Equation of the plane

Given points:

$$P_1(1, 0, 1), \quad P_2(0, 1, 1), \quad P_3(1, 1, 0)$$

Direction vectors:

$$P_1\vec{P}_2 = (-1, 1, 0), \quad P_1\vec{P}_3 = (0, 1, -1)$$

Normal vector:

$$\vec{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = (-1, -1, -1)$$

Using point-normal form:

$$1(x - 1) + 1(y - 0) + 1(z - 1) = 0$$

$$x + y + z = 2$$

Final equation of the plane:

$$\boxed{x + y + z = 2}$$

2) Equation of the line

Points:

$$(1, 2), \quad (3, 1)$$

Direction vector:

$$\vec{v} = (2, -1)$$

Normal vector:

$$\vec{N} = (1, 2)$$

Point-normal form:

$$1(x - 1) + 2(y - 2) = 0$$

$$x + 2y = 5$$

Final equation of the line:

$$\boxed{x + 2y = 5}$$

Distances to Planes and Lines

In this note we look at distances to planes and lines. Our approach is geometric: we draw a sketch and use vector techniques. The sketches are not drawn to scale; they only show the important features of the problem.

1) Distance: Point to Plane

Ingredients:

- A point P
- A plane with normal vector $\vec{N}$ containing a point Q

The distance from P to the plane is

$$d = \left| \vec{PQ} \cdot \vec{N} \right| / \left| \vec{N} \right|.$$

Example:

Let

$$P = (1, 3, 2).$$

Find the distance from P to the plane

$$x + 2y = 3.$$

Answer

First, we gather our ingredients.

$$Q = (3, 0, 0) \quad \text{is a point on the plane}$$

$$\vec{N} = \mathbf{i} + 2\mathbf{j} \quad (\text{normal to the plane})$$

Let R be the point on the plane closest to P (this point is unknown and we do not need to find it).

The figure shows that

$$\text{distance} = |PR| = |\vec{PQ}| \cos \theta = \frac{|\vec{PQ} \cdot \vec{N}|}{|\vec{N}|}$$

Computing

$$\vec{PQ} = 2\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$$

gives

$$\begin{aligned} \text{distance} &= \frac{|\vec{PQ} \cdot \vec{N}|}{|\vec{N}|} = \frac{|(2, -3, -2) \cdot (1, 2, 0)|}{\sqrt{1^2 + 2^2}} \\ &= \frac{|2 - 6|}{\sqrt{5}} = \frac{4}{\sqrt{5}} \end{aligned}$$

2) Distance: Point to Line

Ingredients:

- A point P
- A line with direction vector $\vec{V}$ containing a point Q

The distance from P to the line is

$$d = |QP| \sin \theta = \frac{|\vec{QP} \times \vec{V}|}{|\vec{V}|}$$

We will explain this formula by way of the following example.

Example 2: Let $P = (1, 3, 2)$. Find the distance from the point P to the line through $(1, 0, 0)$ and $(1, 2, 0)$.

Solution:

First, we gather our ingredients.

$$Q = (1, 0, 0)$$

$$\vec{v} = (1, 2, 0) - (1, 0, 0) = (0, 2, 0) = 2\mathbf{j}$$

Let R be the point on the line closest to P .

Using the relation

$$|\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}| \sin \theta$$

the distance is given by

$$\text{distance} = |PR| = |PQ| \sin \theta = \frac{|\vec{QP} \times \vec{v}|}{|\vec{v}|}$$

Now compute

$$\vec{QP} = (1, 3, 2) - (1, 0, 0) = 3\mathbf{j} + 2\mathbf{k}$$

$$\vec{QP} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 2 \\ 0 & 2 & 0 \end{vmatrix} = -4\mathbf{i}$$

$$|\vec{QP} \times \vec{v}| = 4, \quad |\vec{v}| = 2$$

$$\text{Distance} = \frac{4}{2} = 2$$

Distance Between Parallel Planes

The trick here is to reduce the problem to finding the distance from a point to a plane.

Example 3: Find the distance between the planes

$$x + 2y - z = 4 \quad \text{and} \quad x + 2y - z = 1.$$

Both planes have the same normal vector

$$\vec{N} = \mathbf{i} + 2\mathbf{j} - \mathbf{k},$$

so they are parallel.

Take any point on the first plane, say

$$P = (4, 0, 0).$$

The distance between the planes is equal to the distance from point P to the second plane.

Choose a point on the second plane:

$$Q = (1, 0, 0).$$

The distance formula is

$$d = \frac{|\vec{QP} \cdot \vec{N}|}{|\vec{N}|}.$$

Now compute

$$\vec{QP} = (4, 0, 0) - (1, 0, 0) = (3, 0, 0) = 3\mathbf{i}.$$

$$\vec{QP} \cdot \vec{N} = 3\mathbf{i} \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 3.$$

$$|\vec{N}| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}.$$

$$d = \frac{3}{\sqrt{6}} = \frac{\sqrt{6}}{2}.$$

$$\text{Distance between the planes} = \frac{\sqrt{6}}{2}.$$

The planes are as in the previous example. As usual, it is easy to find a point on each plane. Thus, to find the distance between parallel planes, we only need to find the normal vector.

$$\vec{N} = \vec{v}_1 \times \vec{v}_2$$

where $\vec{v}_1$ and $\vec{v}_2$ are the direction vectors of the lines.

Distances to Planes and Lines

1. Distance from a Point to a Plane

Using vector methods, find the distance from the point

$$(1, 0, 0)$$

to the plane

$$2x + y - 2z = 0.$$

2. Distance from a Point to a Line

Using vector methods, find the distance from the point

$$(0, 0)$$

to the line

$$2x + y = 2.$$

Solution:

The distance is given by

$$\text{distance} = |PR| = |PQ| \cos \theta = \frac{|\vec{PQ} \cdot \vec{N}|}{|\vec{N}|}.$$

Let

$$P = (0, 0, 0), \quad Q = (0, 0, 0).$$

Computing

$$\vec{PQ} = \langle 1, 0, 0 \rangle$$

gives

$$\begin{aligned} \text{distance} &= \frac{|\vec{PQ} \cdot \vec{N}|}{|\vec{N}|} = \frac{|\langle 1, 0, 0 \rangle \cdot \langle 2, 1, -2 \rangle|}{3} \\ &= \frac{2}{3} \end{aligned}$$

2.

$$\vec{N} = (2, 1), \quad Q = (1, 0)$$

$$\begin{aligned} \text{distance} &= \frac{|\vec{PQ} \cdot \vec{N}|}{|\vec{N}|} = \frac{|\langle 1, 0 \rangle \cdot \langle 2, 1 \rangle|}{\sqrt{5}} \\ &= \frac{2}{\sqrt{5}} \end{aligned}$$

Solutions of Square Systems

Matrices 3: Homogeneous and Inhomogeneous Systems

Theorems about Homogeneous Systems

On the basis of our work so far, we can formulate a few general results about square systems of linear equations. They are the theorems most frequently referred to in applications.

Definition: The linear system

$$Ax = b$$

is called **homogeneous** if

$$b = 0.$$

Otherwise, it is called **inhomogeneous**.

Theorem 20

If

$$|A| \neq 0,$$

then the system

$$Ax = b$$

has a unique solution given by

$$x = A^{-1}b.$$

Theorem 21

If

$$|A| \neq 0,$$

then the homogeneous system

$$Ax = 0$$

has only the trivial solution

$$x = 0.$$

Note: Theorem (21) is a special case of Theorem (20).